SCHRÖDER PATHS COUNTED BY PEAKS: A FORMAL PROOF OF SCHULTE'S ALTERNATING QUADRATIC IDENTITY (OEIS A060693)

MATH0

Abstract. Let $T(n,k)$ be the number of Schröder paths from $(0,0)$ to $(2n,0)$ with exactly k peaks (OEIS A060693). We prove that $\sum_{k=0}^{n}(-1)^k\,T(n,k)\,(n+1-k)^2 = 1+n+n^2$ for every $n \geq 0$, as conjectured by Werner Schulte in the OEIS entry in 2017, by computing the alternating falling-factorial moments of order $0,1,2$ of the peak statistic, for paths started at any height. The identity is not new: it follows quickly from Harris's formula $T_n(x) = C_n(x+1)$ (2007) relating the row polynomials to the Narayana polynomials, and a Lean proof was merged into the `trureturing` repository in September 2026; ours is an independent formal proof. It is verified in Lean 4 against the frozen, fidelity-reviewed statement that math0 formalized from the OEIS entry, in which $T(n,k)$ is defined by enumerating paths and checked by the kernel against all 56 listed terms.

1. Introduction

A *Schröder path* of semilength n is a lattice path from $(0,0)$ to $(2n,0)$ with steps $U=(1,1)$, $D=(1,-1)$ and $H=(2,0)$ that never goes below the x-axis; a *peak* is a step U immediately followed by a step D. The number $T(n,k)$ of Schröder paths of semilength n with exactly k peaks ($0 \leq k \leq n$) is the triangle A060693 of the OEIS [6], submitted by F. Chapoton in 2001; the path description is a comment of Emeric Deutsch (2003), which became the name of the entry in 2017.

On January 11, 2017, Werner Schulte added to the entry the line

$$\sum_{k=0}^{n}(-1)^k\,T(n,k)\,(n+1-k)^2 = 1+n+n^2, \tag{1}$$

labelled "Conjecture"; two days earlier he had added, without that label, the linear companion $\sum_{k=0}^{n}(-1)^k(1+x(n-k))T(n,k) = x+(1-x)[\,n=0\,]$. In revision #147 of the entry (October 4, 2026) the line (1) is still labelled "Conjecture". We prove it for all $n \geq 0$ (Theorem 1), with a Lean proof of the full statement.

Prior work and status.

- *A short proof from known formulas.* The entry records Mitch Harris's formula $T_n(x) = C_n(x+1)$ (January 2007), where $T_n(x) = \sum_k T(n,k)x^k$ and $C_n(x) = \sum_{k=1}^{n} N(n,k)x^k$, with $N(n,k)$ the Narayana numbers A001263 [5], is the n-th Narayana row polynomial (the formula is meant for $n \geq 1$); equivalently, for all $n \geq 0$, $T(n,k) = \sum_j N(n,j)\binom{j}{k}$ with N the Narayana triangle A090181 [7], which has $N(0,0)=1$ (Ph. Delèham, May 2007). Together with $N(n,1)=1$ and $N(n,2)=\binom{n}{2}$ for $n \geq 1$, this gives (1) in a few lines (Remark 8, not formalized). In substance, the identity is a corollary of formulas that were in the entry before the conjecture was posted.
- *A prior formal proof.* A Lean proof of (1), the theorem `schulte_a060693` in the file `SchroederPeakAlternatingQuadraticMoment.lean`, was merged into the `dev` branch of the GitHub repository `the-omega-institute/trureturing` on September 13, 2026,

Date: October 21, 2026.

in pull request #7436 by ElonSG [1], which supersedes pull request #7402 and was pre-registered in issue #7384. According to the math0 literature referee, who read the source but could not compile it (nor have we), T is defined there through a first-return decomposition of Schröder words, and the proof uses recurrences for the signed falling-factorial moments of orders $0, 1, 2$. It was found by the math0 literature review after our proof was written; the two developments are independent.

- The math0 referees found no other proof in arXiv, MathOverflow or Mathematics Stack Exchange.

We therefore classify our result as an *independent formal proof* of a known result. Our proof uses only the combinatorial definition of T.

2. Statement

The FORMULA section of A060693 [6] (offset 0, $0 \le k \le n$) contains, verbatim, the line

```
Conjecture:  Sum_{k=0..n} (-1)^k*T(n,k)*(n+1-k)^2 = 1+n+n^2.  -
_Werner Schulte_, Jan 11 2017
```

Theorem 1. *For $n, k \ge 0$ let $T(n,k)$ be the number of lattice paths from $(0,0)$ to $(2n,0)$ with steps $U = (1,1)$, $D = (1,-1)$, $H = (2,0)$ that never go below the x-axis and contain exactly k occurrences of U immediately followed by D. Then for every $n \ge 0$,*

$$\sum_{k=0}^{n} (-1)^k \, T(n,k) \, (n+1-k)^2 = 1 + n + n^2.$$

Lean formalization. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A060693.lean`, reviewed for fidelity, and frozen (SHA-256 `7877842d...6b90`) before any proof attempt. Its definitions and the statement read as follows (verbatim; we omit the import, comments, blank lines, all theorems, a `DecidablePred` instance for `IsSchroderPath`, the definitions `valid`, `peaksFrom`, `cnt`, `dTerm`, `step`, `Tab`, `look`, `next`, `tabs`, `Tfast`, `a` of the evaluator and data checks described below, and, at the start of the namespace, the inductive type `Step` with constructors `U`, `D`, `H`, its displacements `Step.dx : Step → ℕ` and `Step.dy : Step → ℤ`, giving $(1,1)$, $(1,-1)$, $(2,0)$, and the line `open Step`):

```
namespace Math0.Statements.A060693
def width (p : List Step) : ℕ := (p.map Step.dx).sum
def height (p : List Step) : ℤ := (p.map Step.dy).sum
def IsSchroderPath (n : ℕ) (p : List Step) : Prop :=
  width p = 2 * n ∧ height p = 0 ∧ ∀ q ∈ p.inits, 0 ≤ height q
def peaks (p : List Step) : ℕ := (p.zip p.tail).count (U, D)
def words : ℕ → List (List Step)
  | 0 => [[]]
  | 1 => [[U], [D]]
  | m + 2 => (words (m + 1)).map (U :: ·) ++ (words (m + 1)).map (D :: ·) ++
      (words m).map (H :: ·)
def schroderPaths (n : ℕ) : Finset (List Step) :=
  (words (2 * n)).toFinset.filter (IsSchroderPath n)
def T (n k : ℕ) : ℕ := ((schroderPaths n).filter (fun p => peaks p = k)).card
def statement : Prop :=
  ∀ n : ℕ, ∑ k ∈ Finset.range (n + 1), (-1 : ℤ) ^ k * (T n k : ℤ) * ((n : ℤ) + 1 - k) ^ 2 =
    1 + (n : ℤ) + (n : ℤ) ^ 2
end Math0.Statements.A060693
```

A path is a word `p : List Step` drawn from the origin. For a word p let $|p| = \#U + \#D + 2\#H$ be its *width* and $\mathrm{ht}(p) = \#U - \#D$ its *height*; after a prefix q the path is at $(|q|, \mathrm{ht}(q))$.

`IsSchroderPath n p` says that the path ends at $(2n, 0)$ and that every prefix (`p.inits` includes the empty prefix and p) ends at height ≥ 0; checking vertices suffices, since the path is straight between them. `peaks p` counts adjacent pairs (U, D). The list `words m` contains every word of width m exactly once (`mem_words`, `nodup_words`), so `T n k` is $T(n, k)$; the file proves `T_eq_ncard`: `T n k = {p : List Step | IsSchroderPath n p ∧ peaks p = k}.ncard`. The sum is computed in $\mathbb{Z}$, so $n + 1 - k$ is not truncated, and `statement` covers all $n \geq 0$ with no further hypothesis.

Data checks. As `T` enumerates exponentially many words, the file also defines a table-based evaluator `Tfast` from the recursion of Lemma 2(b) below and proves `T_eq_Tfast`: `T n k = Tfast n k` for all n, k. Through this lemma, `decide +kernel` checks in the kernel that `T` reproduces rows $n = 0, \ldots, 9$ and $T(10, 0) = 16796$ (`T_data`), and that the triangle read by rows reproduces all 56 terms of the OEIS data (`a_data`). This anchors the definition to the entry independently of the conjecture. The file also checks (1) for $n \leq 12$ (`statement_check`), as a transcription check only.

3. Proof

The proof follows `lean/Math0/Frontier/OEIS/A060693.lean`. Throughout, b ranges over $\{0, 1\}$ (the Lean `false`, `true`), $[\,P\,]$ is 1 if P holds and 0 otherwise, and $\mathcal{A}_m$ is the set of words of width m.

3.1. Generalized counts.

The statement file defines, by recursion on words, `valid` and `peaksFrom`:

$$\begin{aligned}
&\mathrm{valid}_h(\varepsilon) = [\,h = 0\,], \quad \mathrm{valid}_h(Up) = \mathrm{valid}_{h+1}(p), \quad \mathrm{valid}_0(Dp) = 0,\\
&\mathrm{valid}_{h+1}(Dp) = \mathrm{valid}_h(p), \quad \mathrm{valid}_h(Hp) = \mathrm{valid}_h(p);\\
&\mathrm{pk}_b(\varepsilon) = 0, \quad \mathrm{pk}_b(Up) = \mathrm{pk}_1(p), \quad \mathrm{pk}_b(Dp) = b + \mathrm{pk}_0(p), \quad \mathrm{pk}_b(Hp) = \mathrm{pk}_0(p).
\end{aligned}$$

So $\mathrm{valid}_h(p)$ holds if and only if the path p started at height h never goes below the axis and ends on it (`valid_iff`), and $\mathrm{pk}_b(p)$ is the number of peaks of p, plus 1 if $b = 1$ and p begins with D (`peaksFrom_eq`); the flag b records that the step before p was a U. The count `cnt` is

$$c(m, h, b, k) = \#\{p \in \mathcal{A}_m : \ \mathrm{valid}_h(p),\ \mathrm{pk}_b(p) = k\}.$$

Lemma 2 ((a)–(c) in the statement file)**.** *For all $m, h, k, n \in \mathbb{N}$ and $b \in \{0, 1\}$:*

(a) $T(n, k) = c(2n, 0, 0, k)$ *(`T_eq_cnt`);*
(b) $c(m+2, h, b, k) = c(m+1, h+1, 1, k) + \delta(m, h, b, k) + c(m, h, 0, k)$, *where* $\delta = 0$ *if* $h = 0$; $\delta = c(m+1, h-1, 0, k)$ *if* $h \geq 1$, $b = 0$; *and* $\delta = [\,k \geq 1\,]\, c(m+1, h-1, 0, k-1)$ *if* $h \geq 1$, $b = 1$ *(`cnt_step`);*
(c) $c(m, h, b, k) = 0$ *if* $m < h$ *or* $m < k$ *(`cnt_eq_zero`);*
(d) $c(0, 0, b, k) = [\,k = 0\,]$, $c(1, 1, 0, k) = [\,k = 0\,]$ *and* $c(1, 1, 1, k) = [\,k = 1\,]$.

Proof. (a) A word is a Schröder path of semilength n if and only if it has width $2n$ and valid_0 holds, and pk_0 is the number of peaks; as `words (2n)` has no repetitions, counting in the list and in the finset agree. (b) $\mathcal{A}_{m+2} = U\mathcal{A}_{m+1} \sqcup D\mathcal{A}_{m+1} \sqcup H\mathcal{A}_m$, and the recursions above give the contribution of each part; a word Dq is invalid from height 0. (c) A valid word from height h has width $\geq h$ (`valid_le`), and $\mathrm{pk}_b(p) \leq |p|$ (`peaksFrom_le`). (d) $\mathcal{A}_0 = \{\varepsilon\}$ and $\mathcal{A}_1 = \{U, D\}$; from height 1 only D is valid, and $\mathrm{pk}_b(D) = b$ (proof file: `cnt00`, `W_base1f`, `W_base1t`). □

3.2. A bound on the number of peaks.

Lemma 3 (`peaks_bound`)**.** *If* $\mathrm{valid}_h(p)$, *then* $2\,\mathrm{pk}_b(p) + h \leq |p| + 2b$.

Proof. Induction on p. If $p = \varepsilon$, validity forces $h = 0$, and the left side is 0. If $p = Uq$, then $\mathrm{valid}_{h+1}(q)$, and induction with flag 1 gives $2\,\mathrm{pk}_1(q)+h+1 \le |q|+2$, i.e. $2\,\mathrm{pk}_b(p)+h \le |q|+1 = |p|$. If $p = Dq$, then $h \ge 1$ and $\mathrm{valid}_{h-1}(q)$, so $2\,\mathrm{pk}_0(q)+h-1 \le |q|$ and $2\,\mathrm{pk}_b(p)+h = 2b+2\,\mathrm{pk}_0(q)+h \le |p|+2b$. If $p = Hq$, then $2\,\mathrm{pk}_b(p)+h = 2\,\mathrm{pk}_0(q)+h \le |q| < |p|$. □

Corollary 4 (`cnt_peaks`)**.** *If $m < 2k$ then $c(m,0,0,k) = 0$. In particular $T(n,k) = 0$ for $k > n$.*

3.3. **Weighted sums.** For a weight $w\colon \mathbb{N} \to \mathbb{Z}$ put $W_w(m,h,b) = \sum_{k=0}^{m} w(k)\,c(m,h,b,k)$ (Lean `W w m h b`). By Lemma 2(c) the range of summation may be replaced by $\{0,\dots,N-1\}$ for any $N \ge m+1$ (`W_eq_range`); $W_w(m,h,b) = 0$ if $m < h$ (`W_zero_lt`); and W_w is linear in w (`W_lin`). By Lemma 2(d), $W_w(0,0,b) = w(0)$, $W_w(1,1,0) = w(0)$ and $W_w(1,1,1) = w(1)$.

Lemma 5 (`W_step`)**.** *$W_w(m+2,h,b) = W_w(m+1,h+1,1) + W_w(m,h,0) + D_w(m,h,b)$, where $D_w(m,0,b) = 0$, $D_w(m,h+1,0) = W_w(m+1,h,0)$, and $D_w(m,h+1,1) = W_{w(\cdot+1)}(m+1,h,0)$, with $w(\cdot+1)$ the weight $k \mapsto w(k+1)$.*

Proof. Multiply Lemma 2(b) by $w(k)$ and sum over $0 \le k \le m+2$, using `W_eq_range`. For $h \ge 1$, $b = 1$, the sum coming from δ is $\sum_{k=1}^{m+2} w(k)c(m+1,h-1,0,k-1) = \sum_{j=0}^{m+1} w(j+1)c(m+1,h-1,0,j)$. □

3.4. **A closed-form principle.** Valid words from height h have width $m \equiv h \pmod 2$; we write $m = h + 2s$.

Proposition 6 (`closed_gen`)**.** *Let $w\colon \mathbb{N} \to \mathbb{Z}$, $F\colon \mathbb{N}\times\mathbb{N}\times\{0,1\} \to \mathbb{Z}$ and $L\colon \mathbb{N}\times\mathbb{N} \to \mathbb{Z}$ satisfy*

(i) *$F(0,0,b) = w(0)$ for both b, $F(1,0,0) = w(0)$ and $F(1,0,1) = w(1)$;*
(ii) *$W_{w(\cdot+1)}(m+1,h,0) = -W_w(m+1,h,0) + L(h,s)$ whenever $h+2s = m+1$;*
(iii) *for all h,s,b with $h+2s \ge 2$,*

$$\begin{aligned} F(h,s,b) &= [\,s \ge 1\,]\big(F(h+1,s-1,1) + F(h,s-1,0)\big) \\ &\quad + [\,h \ge 1\,]\cdot \begin{cases} F(h-1,s,0) & (b=0),\\ -F(h-1,s,0) + L(h-1,s) & (b=1). \end{cases} \end{aligned} \tag{2}$$

Then $W_w(m,h,b) = F(h,s,b)$ whenever $h+2s = m$.

Proof. Strong induction on m. If $m = 0$ then $h = s = 0$, and if $m = 1$ then $h = 1$, $s = 0$; these cases are (i) with the base values of W_w. Let $m+2 = h+2s$ and apply Lemma 5. If $s = 0$, then $m < h$, so $W_w(m+1,h+1,1) = W_w(m,h,0) = 0$; if $s \ge 1$, induction gives $W_w(m+1,h+1,1) = F(h+1,s-1,1)$ and $W_w(m,h,0) = F(h,s-1,0)$. If $h = 0$ then $D_w(m,h,b) = 0$. If $h \ge 1$ then $(h-1)+2s = m+1$, so by induction $D_w(m,h,0) = F(h-1,s,0)$, and by (ii) and induction $D_w(m,h,1) = -F(h-1,s,0) + L(h-1,s)$. As $h+2s \ge 2$, the total is $F(h,s,b)$ by (2). □

3.5. **The three moments.** The weights $w_0(k) = (-1)^k$, $w_1(k) = (-1)^k k$, $w_2(k) = (-1)^k k(k-1)$ satisfy

$$w_0(k+1) = -w_0(k), \quad w_1(k+1) = -w_1(k) - w_0(k), \quad w_2(k+1) = -w_2(k) - 2w_1(k); \tag{3}$$

let

$$F_0(h,s,b) = [\,s=0\,]\,(1 - 2b[\,h \ge 1\,]), \qquad F_1(h,s,b) = \begin{cases} -b[\,h\ge 1\,] & (s=0),\\ h-1 & (s\ge 1,\ b=1),\\ -(h+1) & (s\ge 1,\ b=0), \end{cases}$$

$$F_2(h,s,b) = [\,s \ge 1\,]\Big((h+1)(s-1)(s+h) + b\big(2h - 2h(s-1)(s+h-1)\big)\Big).$$

Proposition 7 (`W0_closed`, `W1_closed`, `W2_closed`)**.** *For $r = 0, 1, 2$ and all h, s, b: $W_{w_r}(h + 2s, h, b) = F_r(h, s, b)$.*

Proof. Apply Proposition 6 with $(w, F, L) = (w_r, F_r, L_r)$, where $L_0 = 0$, $L_1(h, s) = -F_0(h, s, 0) = -[\, s = 0\,]$ and $L_2(h, s) = -2F_1(h, s, 0) = 2(h + 1)[\, s \geq 1\,]$, in the order $r = 0, 1, 2$.

(ii). By (3) and linearity, $W_{w_0(\cdot+1)} = -W_{w_0}$, $W_{w_1(\cdot+1)} = -W_{w_1} - W_{w_0}$ and $W_{w_2(\cdot+1)} = -W_{w_2} - 2W_{w_1}$. For $r = 1$ (resp. $r = 2$), (ii) then follows from the already proved case $r = 0$ (resp. $r = 1$) at $(m + 1, h, 0)$.

(i). $w_0(0) = 1$, $w_0(1) = -1$, $w_1(0) = 0$, $w_1(1) = -1$, $w_2(0) = w_2(1) = 0$; these are $F_r(0, 0, b)$, $F_r(1, 0, 0)$, $F_r(1, 0, 1)$.

(iii). Let $h + 2s \geq 2$, let R_r be the right side of (2) and $E_r = [\, s \geq 1\,](F_r(h + 1, s - 1, 1) + F_r(h, s - 1, 0))$ its first part. If $s = 0$, then $h \geq 2$ and $E_r = 0$, and $R_0 = 1$ $(b = 0)$, $-1 + 0$ $(b = 1)$; $R_1 = 0$ $(b = 0)$, $-0 + L_1(h - 1, 0) = -1$ $(b = 1)$; $R_2 = 0$; these are $F_r(h, 0, b)$. Let $s \geq 1$. For $r = 0$: $F_0(h, s, b) = 0$, $E_0 = -1 + 1 = 0$ if $s = 1$ and $E_0 = 0$ if $s \geq 2$, and the second part involves only $F_0(h - 1, s, 0) = 0$ and $L_0 = 0$. For $r = 1$: $E_1 = -1$ both for $s = 1$ $(-1 + 0)$ and for $s \geq 2$ $(h - (h + 1))$; for $h \geq 1$ the second part is $F_1(h - 1, s, 0) = -h$ if $b = 0$ and $h + L_1(h - 1, s) = h$ if $b = 1$. So $R_1 = -1 - h$ resp. $-1 + h$, also for $h = 0$, which is $F_1(h, s, b)$. For $r = 2$, $s = 1$: $F_2(h, 1, b) = 2bh$, $E_2 = 0$, and for $h \geq 1$ the second part is $F_2(h - 1, 1, 0) = 0$ if $b = 0$ and $L_2(h - 1, 1) = 2h$ if $b = 1$; so $R_2 = 2bh$. For $r = 2$, $s \geq 2$: put $t = s - 1 \geq 1$. Expanding,

$$E_2 = (h+2)(t-1)(t+h+1)+2(h+1)-2(h+1)(t-1)(t+h)+(h+1)(t-1)(t+h) = t(t+2h+1).$$

With $F_2(h - 1, s, 0) = ht(t + h)$ and $L_2(h - 1, s) = 2h$ we get $R_2 = E_2 + ht(t + h)$ if $b = 0$ and $R_2 = E_2 - ht(t + h) + 2h$ if $b = 1$ (for $h = 0$ both equal E_2). Since $F_2(h, s, b) = (h + 1)t(t + h + 1) + b\,(2h - 2ht(t + h))$, both cases reduce to $t(t + 2h + 1) + ht(t + h) = (h + 1)t(t + h + 1)$, and both sides equal $t(t + ht + h^2 + 2h + 1)$.

In Lean, (iii) is proved by splitting into $s \in \{0, 1, \geq 2\}$, $h \in \{0, 1, \geq 2\}$ and b, and closing the 18 goals for each r by `simp`, followed by `omega` for $r = 0$ and `ring_nf` for $r = 2$. □

3.6. **Proof of Theorem 1.** By Lemma 2(a) the sum in (1) is $\sum_{k=0}^{n}(-1)^k c(2n, 0, 0, k)(n+1-k)^2$. By Corollary 4, $c(2n, 0, 0, k) = 0$ for $n < k \leq 2n$, so we may sum over $0 \leq k \leq 2n$. Since $(n + 1 - k)^2 = (n + 1)^2 - (2n + 1)k + k(k - 1)$, the sum equals

$$(n + 1)^2\, W_{w_0}(2n, 0, 0) - (2n + 1)\, W_{w_1}(2n, 0, 0) + W_{w_2}(2n, 0, 0).$$

By Proposition 7 with $h = 0$, $s = n$, $b = 0$, the moments are $F_0(0, n, 0) = [\, n = 0\,]$, $F_1(0, n, 0) = -[\, n \geq 1\,]$ and $F_2(0, n, 0) = [\, n \geq 1\,](n - 1)n$. For $n = 0$ the sum is 1; for $n \geq 1$ it is $(2n + 1) + n(n - 1) = n^2 + n + 1$. □

Remark 8 (the Narayana route; not formalized)**.** Let $N(n, j)$ be the number of Dyck paths of semilength n with j peaks, so $N(0, 0) = 1$, and for $n \geq 1$, $N(n, 0) = 0$, $N(n, 1) = 1$ (only $U^n D^n$) and $N(n, 2) = \binom{n}{2}$ [5, 7]. Replacing every H of a Schröder path by UD gives a Dyck path whose peaks are the old peaks and the replaced steps; conversely, flattening any $j - k$ of the j peaks of a Dyck path into H steps creates no new peak. Hence $T(n, k) = \sum_j N(n, j)\binom{j}{k}$, i.e. $T_n(x) = \sum_j N(n, j)(1 + x)^j$ (Harris's formula). Since $\sum_k (-1)^k \binom{j}{k} k^{\underline{r}} = (-1)^r r!\,[\, j = r\,]$, we get $\sum_k (-1)^k k^{\underline{r}} T(n, k) = (-1)^r r!\, N(n, r)$, which for $n \geq 1$ gives the moments 0, -1, $n(n - 1)$ found above.

4. Machine verification

Files and declarations. The proof is in `lean/Math0/Frontier/OEIS/A060693.lean` in the math0 repository, `https://github.com/Foundation-0/math0`. It imports the frozen statement

file `lean/Math0/Statements/OEIS/A060693.lean` and the audit commands in `lean/Math0/Audit.lean`. The main declaration is `Math0.Frontier.A060693.result : statement`, followed by the command

```
#math0_settles Math0.Frontier.A060693.result Math0.Statements.A060693.statement
```

All auxiliary declarations are private; their names are given with the corresponding results in Section 3, and parts (a)–(c) of Lemma 2 are proved in the statement file. Neither file uses `sorry`, `native_decide`, custom axioms or `set_option`. The proof file uses no `decide`; the only kernel evaluations are the data checks of the statement file, which the proof does not use.

Audit. The command `#math0_settles` (in `lean/Math0/Audit.lean`, for statements that math0 formalizes itself) fails unless `statement` is a `Prop`-valued definition without universe parameters; `statement` and `result` depend on no axioms besides `propext`, `Classical.choice` and `Quot.sound` (this rejects `sorryAx`, `Lean.ofReduceBool` from `native_decide`, and custom axioms); and the type of `result` is syntactically `statement` or $\neg$ `statement`, up to metadata, with no unfolding. Here it reports

```
math0 audit: 'Math0.Statements.A060693.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A060693.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A060693.result' proves 'Math0.Statements.A060693.statement'
```

Toolchain and reproduction. We use Lean 4 [4] `v4.33.1` and Mathlib [3] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [2] at commit `c77fdd4`, which this result does not use. To reproduce, run `scripts/verify.sh` from the repository root: it fetches the Mathlib cache, builds all math0 results with `lake build` (re-running every audit command) and checks that the pinned `formal_conjectures` and `mathlib` checkouts are unmodified. The single file can be checked with `lake env lean Math0/Frontier/OEIS/A060693.lean` in `lean/`; on our machine this took under a minute, including loading Mathlib.

References

[1] ElonSG, *prove: OEIS A060693 Schröder-path peak identity (Schulte 2017) — v2: public peak law + Scribe repairs*, pull request #7436 to `the-omega-institute/trureturing` (branch `dev`), merged September 13, 2026, `https://github.com/the-omega-institute/trureturing/pull/7436`; supersedes pull request #7402; pre-registration issue #7384.

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